theorem; its clear paradigmatic simplicity as a kind of parallel computation. There is no reason to suppose that any of these virtues carry over to the many-layered version. Nevertheless, we consider it to be an important research problem to elucidate (or reject) our intuitive judgment that the extension is sterile. Perhaps some powerful convergence theorem will be discovered, or some profound reason for the failure to produce an interesting "learning theorem" for the multilayered machine will be found.

13.3 Analyzing Real-World Scenes

One can understand why you, as mathematicians, would be interested in such clear and simple predicates as ψ_{PARITY} *and* $\psi_{\text{CONNECTED}}$. *But what if one wants to build machines to recognize chairs and tables or people? Do your abstract predicates have any relevance to such problems, and does the theory of the simple perceptron have any relevance to the more complex machines one would use in practice?*

This is a little like asking whether the theory of linear circuits has relevance to the design of television sets. Absolutely, some concept of connectedness is required for analyzing a scene with many objects in it. For the whole is just the sum of its parts and their interrelations, and one needs some analysis related to connectedness to divide the scene into the kinds of parts that correspond to physically continuous objects.

Then must we conclude from the negative results of Chapter 5 that it will be very hard to make machines to analyze scenes?

Only if you confine yourself to perceptrons. The results of Chapter 9 show that connectivity is not particularly hard for more serial machines.

But even granting machines that handle connectivity, isn't there an enormous gap from that to machines capable of finding the objects in a picture of a three-dimensional scene?

The gap is not so large as one might think. To explain this, we will describe some details of a kind of computer program that can do it. The methods we will describe fall into the area that is today called "heuristic programming" or "artificial intelligence."*

*See, for example, *Computers and Thought* (Feigenbaum and Feldman, 1963) and *Semantic Information Processing* (Minsky, 1968).